

EE 2101: Quiz 1

1. Consider an LTI system with the transfer function

[Marks: 3]

$$T(s) = \frac{1}{(s+1)(s+4)^2}$$

Find the unit step response of this system.

2. Consider a hypothetical LTI system whose output $y(t)$ corresponding to impulse input is: [Marks: 3]

$$y(t) = \begin{cases} -1, & \text{if } t \in [2k, 2k+1) \\ +1, & \text{if } t \in [2k+1, 2k+2) \end{cases}$$

for $k = 0, 1, 2, 3, \dots$. Determine whether this system is BIBO stable or not. Justify your answer.

3. Consider a second-order system whose unit step response is as shown in Figure 1. Find the transfer function of the system. [Marks: 3]

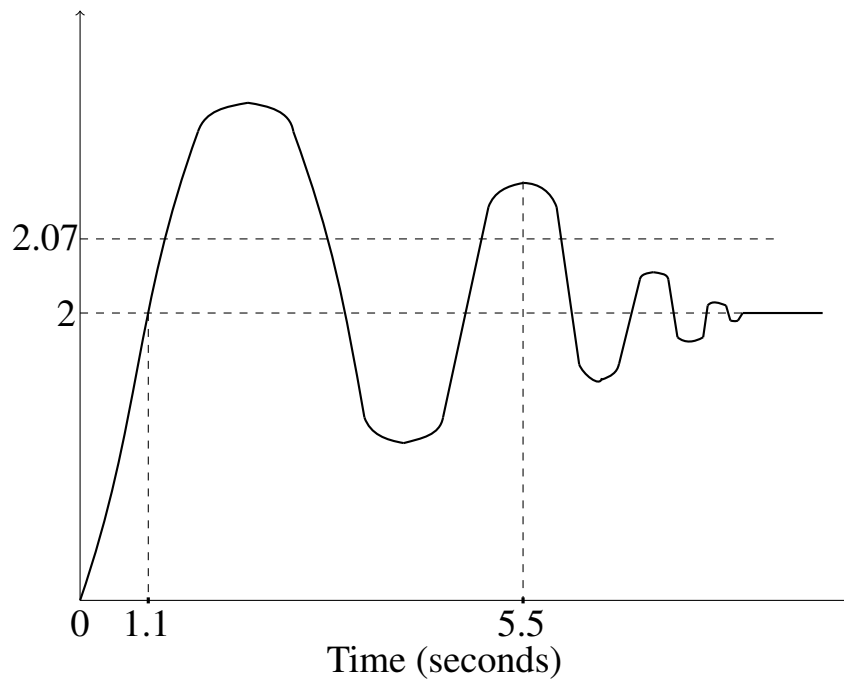


Figure 1: Unit Step Response

4. Consider control system shown in Figure 2 with

[Marks: 4]

$$G(s) = \frac{K(s+3)(s+5)}{(s-1)(s-2)(s-4)(s-9)}$$

Find the values of K at which system has closed-loop poles on the imaginary axis. Also find those poles.

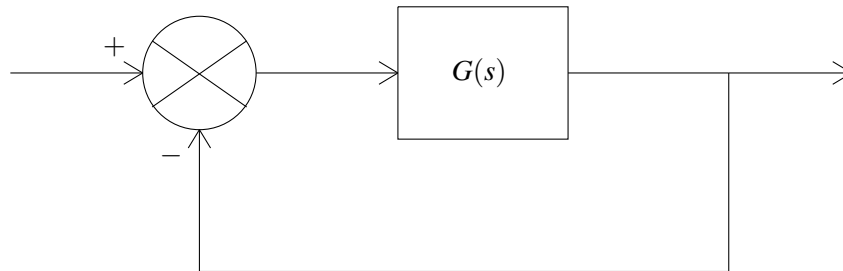


Figure 2: Control System

5. Consider a second order LTI system, say S_1 , with settling time 4 seconds. Let S_2 be another LTI system with transfer function

[Marks: 7]

$$T(s) = \frac{1}{s^5 + 20s^4 + 150s^3 + 530s^2 + 989s + 1110}$$

It is given that systems S_1 and S_2 have a common pole. Find the settling time of system S_2 .